

Core Relations of the Universal Quantum Foam Hypothesis (UQSH) A Minimal Set of Formal Interfaces

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Abstract

The Universal Quantum Foam Hypothesis (UQSH) proposes a continuous field ontology in which physical phenomena emerge from different organizational regimes of a single dynamical medium. This document collects the minimal mathematical relations that serve as formal interfaces between the ontological assumptions of the UQSH and established physical descriptions. It is not intended as a complete field theory, but as a compact reference enabling consistency checks, further formal development, and comparison with observational frameworks. All relations are stated in their minimal form, with explicit indication of their domain of applicability and open parameters.

1 Field State and Basic Assumptions

The physical state of space is described by a scalar field quantity

$$Qu(x, t) \tag{1}$$

denoting the local tension or excitation state of the underlying medium. For quasistatic or weak-field considerations, the time dependence may be suppressed and the reduced notation $Qu(x)$ is used.

The UQSH assumes:

- a single continuous field medium,
- universal reorganization dynamics,
- no fundamental distinction between “matter”, “energy”, and “geometry” beyond organizational state.

2 Effective Gravitational Potential (Quasistatic Regime)

In the weak-field, quasistatic limit, the classical gravitational potential is identified with the local field state:

$$\Phi(x) = \alpha Qu(x), \quad (2)$$

where α is a proportionality factor depending on units and normalization conventions. The observable acceleration follows as

$$\vec{a}(x) = -\nabla\Phi(x) = -\alpha\nabla Qu(x). \quad (3)$$

This relation expresses gravitation as the local reorganization response of the field to spatial inhomogeneities of its tension state.

3 Source Term and Bound Field Organization

In this framework, the source of the effective gravitational field is not a substantial mass density, but the density of bound field organization, denoted $\rho_{bind}(x)$. In the weak-field regime, a Poisson-type relation applies:

$$\nabla^2 Qu(x) \propto \rho_{bind}(x). \quad (4)$$

The quantity $\rho_{bind}(x)$ characterizes the locally stabilized, non-freely propagating fraction of field organization.

4 Minimal Models for the Binding Density

To avoid arbitrariness, ρ_{bind} is taken to depend on the field state itself. Two minimal forms are proposed as working models:

Local binding model

$$\rho_{bind}(x) = \kappa f(Qu(x)), \quad (5)$$

with f a monotonic function.

Structure-sensitive binding model

$$\rho_{bind}(x) = \kappa \left(\frac{1}{2} K Qu(x)^2 + \frac{1}{2} L |\nabla Qu(x)|^2 \right), \quad (6)$$

where the first term measures local field tension and the second term captures stored energy in spatial organization. Here $|\nabla Qu(x)|^2$ denotes the squared norm of the spatial gradient. The constants κ, K, L remain to be fixed by physical calibration or further theoretical constraints.

5 Energy, Mass, and Inertia

Within the UQSH, energy and mass do not represent independent substances, but measures of field organization.

- Energy quantifies the degree of field excitation or reorganization.
- Mass corresponds to the degree of stabilized (bound) organization and manifests as inertia.

The relation $E = mc^2$ is interpreted as a statement of equivalence between bound field energy and inertial resistance, consistent with the UQSH ontology.

6 Regime Limits: Horizon and Void

Two asymptotic organizational regimes are distinguished:

- $Qu \rightarrow Qu_{Horizon}$: maximally bound organization with predominantly internalized dynamics.

- $Qu \rightarrow Qu_{Void}$: minimally bound organization with predominantly free propagation.

Neither regime represents a termination of dynamics; both retain residual reorganization. They are interpreted as limiting forms of the same underlying field behavior.

7 Scale-Connection Criterion

Transitions between descriptive scales are characterized by dimensionless invariants rather than absolute lengths. A minimal local scale indicator is given by

$$\epsilon(L) = \frac{L|\nabla Qu|}{|Qu|} \quad (7)$$

An effective connection scale L^* is defined implicitly by $\epsilon(L^*) \approx 1$, marking the transition between locally homogeneous and structurally dominated descriptions.

8 Scope and Status

This document provides a compact set of formal relations serving as interfaces between the UQSH ontology and standard physical frameworks. It does not claim completeness and introduces no fixed numerical predictions. Its purpose is to define a common reference point for consistency analysis, further formalization, and comparison with observational phenomena.

- Parts of the editorial preparation were carried out with the assistance of AI-based tools.

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